0:00

0 seconds

Let's talk about liver failure. I was trying to This is my second time having to do this, so I'm just deleting some

0:06

6 seconds

stuff because my preview is rough when it comes to um marking things up.

0:13

13 seconds

Anyways, so cerosis is also known as liver failure. We're first going to talk about the patho physiology of it, which is in purple here, these four steps, and

0:21

21 seconds

then we're going to do like our normal liver functions and then how they are all messed up because of liver failure or cerosis. So,

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29 seconds

Let me show you a couple slides that I have from my adult health PowerPoint.

0:35

35 seconds

Um, when we talk about something that causes damage to the liver, mostly I see alcohol or that's just my acronym for

0:43

43 seconds

alcohol. Uh, it's pretty widespread acronym though. Anyways, um,

0:52

52 seconds

it can not only just be alcohol, but it can be biliary problems, hepattoxins, hepatitis B and C. We'll talk about hepatitis in another video, but

1:01

1 minute, 1 second

something causes liver damage. So, let me tell you how biliary problems will cause liver damage. So, normally our

1:10

1 minute, 10 seconds

liver makes bile and we need bile to um digest fats. So, the bile travels

1:19

1 minute, 19 seconds

down these ducts, these green ducts, and into the dadum.

1:25

1 minute, 25 seconds

This is the first part of our small intestine and this is where bile gets secreted into so that we can digest fats from our intestine.

1:36

1 minute, 36 seconds

The gallbladder is just a storage point for bile. So the gallbladder does

1:44

1 minute, 44 seconds

nothing except store bile. Okay? The liver is what makes it. The gallbladder stores it. And then you also have these

1:51

1 minute, 51 seconds

ducts that run through the pancreas and they're responsible for secretreting pancreatic enzymes. So these are enzymes

1:58

1 minute, 58 seconds

that help break down other nutrients um so that we can absorb them in the datadum. So they all go through these this duct system.

2:10

2 minutes, 10 seconds

So if we have biliary problems like bile biliary um then this like let's say we have a

2:19

2 minutes, 19 seconds

stone sitting here let's see if I can mark this up draw okay let's say I have like a gall stone

2:26

2 minutes, 26 seconds

sitting here okay so none of this bile

2:34

2 minutes, 34 seconds

can get past the stone kind of like a kidney stone we talked about so all this bile is going to back up and it's going

2:40

2 minutes, 40 seconds

to cause congestion in the liver and it's going to cause liver damage that way.

2:49

2 minutes, 49 seconds

Um, I think that's all I want to mention there. I do just want to quickly go over this though because I think this is a

2:56

2 minutes, 56 seconds

good picture of the stages of liver damage. So when we talk about liver failure or cerosis,

3:05

3 minutes, 5 seconds

we are here. Okay. So, we have a healthy liver.

3:10

3 minutes, 10 seconds

Our liver doesn't like a whole lot of well, it's really good at detoxing stuff. We'll talk about that. But when it doesn't like a whole lot of stuff, it

3:19

3 minutes, 19 seconds

gets fatty and then the fat cells it it can't like yeah our liver can like regenerate tissue.

3:29

3 minutes, 29 seconds

But if we keep putting bad stuff in our liver that our liver doesn't like and it turns like cells into fat,

3:38

3 minutes, 38 seconds

then over time our liver just starts to scar over. like instead of trying to heal this fatty tissue, our liver will

3:46

3 minutes, 46 seconds

just scar. So that's what we call liver fibrosis. So remember fibrosis equals scar tissue.

3:53

3 minutes, 53 seconds

Cerosis is just like widespread scar tissue, fibrosis like it's it's the entire liver. I will show you a good

4:01

4 minutes, 1 second

picture. Oh, here's a fatty liver. Okay, normal liver, fatty liver. And then here's our cerosis. Okay.

4:10

4 minutes, 10 seconds

So, it's got all these like lobule things in it and it's all the scar tissue.

4:20

4 minutes, 20 seconds

So, when we think about the pathophysiology of cerosis and we have a bunch of scar tissue because our liver is fibrodic.

4:29

4 minutes, 29 seconds

We learned last week that our skin and our cells once they develop scar tissue and

4:37

4 minutes, 37 seconds

fibrosis that they do not perform the same function as normal cells in that area. So like our skin cells don't have

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4 minutes, 44 seconds

hair. Uh they don't sweat. Same with our liver. Our liver can't do the things that it's supposed to do when there's

4:51

4 minutes, 51 seconds

scar tissue there. Also, scar tissue is tough, way tougher tissue than normal um liver

4:59

4 minutes, 59 seconds

tissue. So, if we need if our liver takes a lot of blood flow to it, the scar tissue is going to pretty much

5:07

5 minutes, 7 seconds

prevent that blood flow. It is harder to get blood flow to scar tissue or through scar tissue than regular tissue, which

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5 minutes, 15 seconds

we also learned. So that is the basis for our um cerosis.

5:23

5 minutes, 23 seconds

So something causes damage. We've got inflammation that progresses and causes

5:29

5 minutes, 29 seconds

necrosis. So that's kind of like that um fatty liver stage. We start to get some

5:37

5 minutes, 37 seconds

some cell death over time unable to regenerate. It turns into permanent liver damage and fibrosis and scar tissue. So with that fibrosis and scar

5:47

5 minutes, 47 seconds

tissue, our liver can't function normally. So pretty much what I just explained. So let's look at our normal

5:54

5 minutes, 54 seconds

liver functions here. Okay, so we talked already it makes bile.

6:01

6 minutes, 1 second

Okay, makes bile. It removes and conjugates Billy Rubin.

6:07

6 minutes, 7 seconds

So conjugate Billy Rubin means

6:14

6 minutes, 14 seconds

if we are removing Billy Rubin from our body that's one thing but conjugation

6:22

6 minutes, 22 seconds

means that the Billy Rubin we're not only going to get rid of some but we're also going to conjugate or put some

6:32

6 minutes, 32 seconds

in vile and we're going to get rid of it that Okay. So, we can remove it directly. We can just get rid of it or

6:41

6 minutes, 41 seconds

we can conjugate it or put some in the bile.

6:45

6 minutes, 45 seconds

Um, our liver helps make clotting factors. Clotting factors are part of the clotting cascade and clotting system to help clot blood when we are bleeding.

6:56

6 minutes, 56 seconds

Protein synthesis so makes proteins.

6:59

6 minutes, 59 seconds

Glycogen storage. Glycogen is that stored form of glucose. So, our body has extra glucose. stored in our liver as

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7 minutes, 7 seconds

glycogen to be released when we are hypoglycemic or have low blood sugar.

7:13

7 minutes, 13 seconds

Our um liver will break down the glycogen and release it into the blood to help maintain glucose levels.

7:23

7 minutes, 23 seconds

Um it stores iron and vitamin B12. Those are very important when we talk about anemia, estrogen and eldoststerone inactivation.

7:34

7 minutes, 34 seconds

So, our liver functions to deactivate or inactivate estrogen and aldoststerone. Remember, aldoststerone

7:42

7 minutes, 42 seconds

is sodium and water. Let's see if I can do it. Oh, it's not going to let me.

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7 minutes, 49 seconds

Oh my gosh, I got so frustrated this.

7:52

7 minutes, 52 seconds

Okay, water reabsorption and excretion of potassium.

7:57

7 minutes, 57 seconds

Detoxes remove harmful substances and lots and lots. That's I'll move it down there. Lots and lots of blood flow. Tons

8:06

8 minutes, 6 seconds

of blood flow through my liver. So, let's talk about each of these.

8:11

8 minutes, 11 seconds

Okay. If my liver can't make bile, well, what is bile for?

8:16

8 minutes, 16 seconds

Bile is to help digest the fats. It's released in the dadum by the the duct, the bile ducts to help digest fat in our

8:24

8 minutes, 24 seconds

intestines. So, if I can't make the bile, then I can't digest fat. And I can't digest the fat soluble vitamins A, D, E, and K.

8:33

8 minutes, 33 seconds

And I can't absorb them. Digest, absorb, whatever you want to say with the fat soluble vitamins. Okay? So that causes low levels of vitamins A, D, E, and K.

8:45

8 minutes, 45 seconds

And it causes something called stattoriia.

8:49

8 minutes, 49 seconds

So stayataterhea is fatty stools. Sorry, I can't like cross my tees or dot my eyes because this thing's

8:57

8 minutes, 57 seconds

going to go whack maybe. Okay, so fatty stools. So, if I can't digest

9:04

9 minutes, 4 seconds

fat and I can't absorb the fats that I'm eating, I have to get rid of it somehow.

9:10

9 minutes, 10 seconds

And if it's it it's found in my intestine, like I eat, fats are in my intestine. I don't have the bile to

9:18

9 minutes, 18 seconds

digest it. So, it's going to stay in my intestines and it's just going to come out and poop. So it's fatty stools the

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9 minutes, 26 seconds

stateria they're usually very foul smelling um

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9 minutes, 33 seconds

we talk about more when we talk about um diarrhea and stuff but that's staturia.

9:41

9 minutes, 41 seconds

Okay. How about the clotting factors? So if we can't make the clotting factors then we're going to bleed more. And if

9:49

9 minutes, 49 seconds

we bleed more, we're going to have signs like petiki, those like dot hemorrhages on the skin. They're like little red

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9 minutes, 55 seconds

dots. And echimosis. Echimosis is bruising.

10:03

10 minutes, 3 seconds

I put some labs here for you guys so you guys can start getting used to some labs. So clotting factors, the main ones

10:10

10 minutes, 10 seconds

you will see are PT INR, PTT, and platelets.

10:15

10 minutes, 15 seconds

So PT, INR, and PTT are all about the time

10:23

10 minutes, 23 seconds

it takes to oh my gosh, time it takes to clot.

10:36

10 minutes, 36 seconds

So what these levels are saying is that it's taking longer time or increased time to clot.

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10 minutes, 44 seconds

which would mean I'm bleeding more. Like if it's taking if I have a cut on my arm and it's bleeding, the longer it bleeds is the longer it takes them it to clot.

10:57

10 minutes, 57 seconds

So we have increased clotting times. PT INR PTT.

11:02

11 minutes, 2 seconds

Platelets. Platelets help they're like the sticky cells that help clot off um areas that are bleeding or damaged.

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11 minutes, 13 seconds

So if we have low platelets and not a whole lot of those sticky things to help clump together and stop

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11 minutes, 21 seconds

bleeding, then we're going to bleed more.

11:24

11 minutes, 24 seconds

Okay? So decrease platelets puts us at risk for bleeding.

11:29

11 minutes, 29 seconds

All right? Billy Rubin, we can't remove it or conjugate it. So we can't get rid of it and we can't put it in the bile.

11:36

11 minutes, 36 seconds

And we're not making bile. So our body, this Billy Ribbon has to go somewhere.

11:43

11 minutes, 43 seconds

And we get rid of it. If we can't get rid of it by our liver and in our bile, then it's going to go one of two places.

11:52

11 minutes, 52 seconds

It's either going to go in our blood, and that's how we get increased Billy Rubin levels in our blood or it's got to be excreted by the kidneys.

12:00

12 minutes

Okay, remember the bile goes in those ducts.

12:05

12 minutes, 5 seconds

Let me put this up again. Okay, if I cannot make bile and I can't put Billy

12:12

12 minutes, 12 seconds

Rubin in the bile, then it's not going in the intestine.

12:18

12 minutes, 18 seconds

It's staying up here. Billy Rubin is staying up here. So, it's going to go in the blood

12:27

12 minutes, 27 seconds

or it's going to be excreted by the kidneys and urine.

12:34

12 minutes, 34 seconds

So when it's when we have high Billy Rubins and it's in the a ton of it's in the blood

12:43

12 minutes, 43 seconds

they have we call it like patients have jaundice. Let me show you a picture of jaundice.

12:50

12 minutes, 50 seconds

This is jaundice. This yellowing of the eyes. This man is very yellow. Okay.

12:55

12 minutes, 55 seconds

Billy Rubin causes yellowing of the skin. Another term you might see is ectaric.

13:02

13 minutes, 2 seconds

That also means jaundice. Okay. So, ectaric yellowing.

13:09

13 minutes, 9 seconds

That's pretty bad. I've seen that bad of jaundice. It does exist. It's very sad.

13:16

13 minutes, 16 seconds

Um, and Billy Rubin is also very itchy. So, it kind of like so it makes that yellow

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13 minutes, 24 seconds

color of our skin. and it kind of deposits in our skin and causes us to itch.

13:30

13 minutes, 30 seconds

When it's excreting the kidneys, you are going to see dark urine, dark dark dark urine because Billy Rubin is like that

13:38

13 minutes, 38 seconds

dark yellow color. So, it's going to make our urine turn dark. And it's not it it when Billy Rubin um and bile

13:48

13 minutes, 48 seconds

function normally, it gives us it gives our stool or poop that nice like brown color.

13:57

13 minutes, 57 seconds

And so, when we don't have that in the intestines, it we're going to have clay colored stools because we don't have any Billy Rubin going into our intestines.

14:07

14 minutes, 7 seconds

All the belly rubin we have is going into our blood and and being excreted by our kidneys.

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14 minutes, 14 seconds

Proteins. Okay. So, if we can't make proteins in our liver, then we're going to have decreased albumin.

14:22

14 minutes, 22 seconds

Low levels of albammen. If we have low levels of albumin in our body, in our vessels,

14:31

14 minutes, 31 seconds

let's see. This is okay. It's not going to let me draw P. That's fine. I was going to put P for protein, but lines work too. Okay. So, when we have low

14:41

14 minutes, 41 seconds

levels of albmen or protein in our blood, our

14:49

14 minutes, 49 seconds

fluid in our vessels likes to go where there's more protein.

14:56

14 minutes, 56 seconds

And I have that decreased. I shouldn't really explain it like that. It's more of that pull pressure. Let's just say

15:04

15 minutes, 4 seconds

that this protein shouldn't be out here. I don't like the way I explain that. Okay.

15:11

15 minutes, 11 seconds

If I have all this fluid in here, all this fluid out here in my interstitial space, if I have low protein in my vessel, then

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15 minutes, 20 seconds

I'm not pulling as much fluid back in to my vessels or my vascule.

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15 minutes, 27 seconds

So, this can result in edema and two different kinds. So ascites is that third spacing.

15:35

15 minutes, 35 seconds

It's in that space where we cannot absorb it again. So ascites, here's your picture for ascites right here. This big

15:41

15 minutes, 41 seconds

old belly. Okay. This is all fluid in the peritineal cavity. Okay. It looks like this patient is pregnant. The other

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15 minutes, 50 seconds

thing you will notice is these like very distended abdominal veins. Okay.

15:58

15 minutes, 58 seconds

So they get distended abdominal veins.

16:01

16 minutes, 1 second

and all this fluid buildup in their peritineal cavity that is ascites.

16:07

16 minutes, 7 seconds

The other thing that can happen to them is peripheral edema. So that this edema just doesn't have to relate to third spacing. It can also just be peripheral

16:15

16 minutes, 15 seconds

edema. If I have peripheral edema, I have increased risk for skin infection because

16:23

16 minutes, 23 seconds

my tissue is more likely to break down when there's like a layer of water between my epithelium and um my muscle and my vessels.

16:35

16 minutes, 35 seconds

If I have a sites, look at this. Look at this poor person's belly. Look how big it is. For those of you that have been pregnant before, um, think of how well

16:44

16 minutes, 44 seconds

you could breathe when you were pregnant. Not very well, especially when you're like third trimester. So that's

16:50

16 minutes, 50 seconds

these patients are like third trimester most of the time. So if we can't fully expand our lungs, we are increased for respiratory infections.

17:05

17 minutes, 5 seconds

All right, let's do estrogen. So if we cannot deactivate or inactivate estrogen, we're going to have increased

17:12

17 minutes, 12 seconds

estrogen levels in our body and that causes spider nevi gynecomastia and irregular menes. So spider nevi are

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17 minutes, 20 seconds

these this is called palmer aiththemma.

17:28

17 minutes, 28 seconds

Um that's something else that can happen with increased levels of estrogen.

17:32

17 minutes, 32 seconds

Gynecomastia is increased breast tissue and irregular menes is just irregular periods.

17:41

17 minutes, 41 seconds

Aldoststerone if we cannot inactivate eldoststerone then we have increased levels of aldoststerone in our body

17:48

17 minutes, 48 seconds

because we can't like deactivate it. We can't stop it.

17:53

17 minutes, 53 seconds

So if we have increased eldoststerone that means we're retaining more sodium and water. If we retain more sodium and

18:00

18 minutes

water in our vessels, we have increased blood volume or hypervalmia. That increased hydrostatic pressure. We have increased hydrostatic

18:09

18 minutes, 9 seconds

pressure is also is also a risk factor for edema. Oh my gosh.

18:19

18 minutes, 19 seconds

Okay, I can't draw a circle. This thing is being crazy. All right, glycogen.

18:29

18 minutes, 29 seconds

So, we can't store glycogen. If our liver normally stores glycogen and breaks glycogen down when our blood sugar is low, we can't store it and we

18:37

18 minutes, 37 seconds

can't use it. Okay? So, glycogenolyis is that breakdown of glycogen.

18:42

18 minutes, 42 seconds

So, if we can't store glycogen, as soon as we use up our sugar or glucose in our body, we are hypoglycemic and there's no

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18 minutes, 50 seconds

compensatory mechanism for it. So decrease blood sugar, iron and B12, we can't store it, so we

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18 minutes, 58 seconds

get anemia. We need iron and B12 to help make red blood cells or hemoglobin.

19:06

19 minutes, 6 seconds

And if we can't do that, then we have anemia.

19:11

19 minutes, 11 seconds

The detoxification and removal of harmful substances. So if our liver normally can remove harmful

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19 minutes, 19 seconds

substances then or toxins let's say they are toxins our toxins are going to build up. The

19:27

19 minutes, 27 seconds

main one that they get all the time is ammonia.

19:32

19 minutes, 32 seconds

Ammonia is a end product of protein metabolism.

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19 minutes, 40 seconds

So the liver turns ammonia into ura which is excreted by the kidneys and that's how we get our bun our blood ura nitrogen.

19:51

19 minutes, 51 seconds

So byproduct of protein metabolism.

19:55

19 minutes, 55 seconds

Okay. So we get ammonia and our ammonia our liver cannot break this down into that ura. So we get high levels of

20:03

20 minutes, 3 seconds

ammonia which cause something called hpatic encphylopathy.

20:08

20 minutes, 8 seconds

Hypatic encphylopathy is basically confusion, disorientation

20:17

20 minutes, 17 seconds

like weird neuros signs and symptoms that you get from specifically high ammonia levels. So some of the things so

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20 minutes, 25 seconds

that you know hpatic and sephylopathic patients are going to present with confusion. seizures can have coma.

20:32

20 minutes, 32 seconds

Coma, fancy term for basically unresponsive.

20:36

20 minutes, 36 seconds

Um, or asterexis. And asterexus is if you ask the patient to put out their arms.

20:44

20 minutes, 44 seconds

Oh my gosh, this thing. Okay, I guess I can't write. It's like a hand flapping. It's like a hand flapping tremor.

20:50

20 minutes, 50 seconds

Another thing they do too is like um they like have weird tongue movements and kind of like move their head side to side.

21:00

21 minutes

Um I've seen pretty bad hypatic and sephilopathy. They're very very confused.

21:08

21 minutes, 8 seconds

So the last thing here is blood flow.

21:12

21 minutes, 12 seconds

And this blood flow issue causes a lot of our issues in cerosis. So, our liver

21:20

21 minutes, 20 seconds

has tons of blood going through it. Tons and tons of blood.

21:26

21 minutes, 26 seconds

So, let me pull up another picture for you. Okay, here's our portal system.

21:34

21 minutes, 34 seconds

Okay, so what our liver does, the reason there's so much blood going through our liver is because it's supplied by a hypatic artery.

21:46

21 minutes, 46 seconds

atic artery that supplies oxygenated blood to the liver so the liver can function. And it

21:54

21 minutes, 54 seconds

also gets fed blood by the portal system. And the portal system is

22:00

22 minutes

composed of all these veins that take nutrients that are absorbed from the

22:07

22 minutes, 7 seconds

intestine, the stomach, the spleen, and they put it in the liver so that the liver can then use and like kind of like

22:17

22 minutes, 17 seconds

go through all the nutrients that have been absorbed before the nutrients get put into the systemic circulation.

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22 minutes, 29 seconds

into the systemic venus system via the inferior vennea. Okay.

22:36

22 minutes, 36 seconds

So this allows this cool system that we have all these veins

22:42

22 minutes, 42 seconds

from our stomach, intestine, spleen, they take these absorbed nutrients, put it in the liver and our liver can kind of like detox things that way.

22:55

22 minutes, 55 seconds

So this causes a lot of issues though. When our liver is looking real gross and

23:03

23 minutes, 3 seconds

serotic and bumpy and we've got all this scar tissue in here, it doesn't make blood flow through this organ very easy.

23:12

23 minutes, 12 seconds

It is hard to get blood flow to go through scar tissue.

23:16

23 minutes, 16 seconds

So, what happens is, let me Oh, I can't Oh, there we go. Okay, I'm

23:25

23 minutes, 25 seconds

gonna erase this arrow. What happens is all this blood flow, it's trying to like get all these nutrients back to the

23:33

23 minutes, 33 seconds

liver to go through it and sort it all out, but the scar tissue is kind of preventing that blood flow. So, it all backs up.

23:45

23 minutes, 45 seconds

Okay? All this blood flow backs up and that is what we call portal hypertension.

23:54

23 minutes, 54 seconds

So we've got increased pressure in our portal veins.

24:01

24 minutes, 1 second

This causes a slew of issues.

24:04

24 minutes, 4 seconds

Okay, let's first look at what the portal veins connect to. So it connects to

24:13

24 minutes, 13 seconds

our intestine, stomach, kind of first part of us of our esophagus.

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24 minutes, 20 seconds

When we have dilated and engorged veins with lots of pressure in them, we call them varices like varicose veins.

24:31

24 minutes, 31 seconds

So this is how they get esophageal veraces.

24:35

24 minutes, 35 seconds

Oh, our body also like in response to this hypertension, it creates collaterals. Collaterals are a fancy

24:43

24 minutes, 43 seconds

term for like extra veins to try to like spread out the blood flow and decrease some of the pressure. So we got all

24:50

24 minutes, 50 seconds

these veins that like travel up and out way more than the normal person has. So

24:58

24 minutes, 58 seconds

this increased blood flow in all these different areas get engorged and dilated and we call those veraces. Veraces

25:07

25 minutes, 7 seconds

these veins have very weak very weak walls and under such high pressure like Venus is Venus blood is not supposed to

25:16

25 minutes, 16 seconds

have such high pressure but because all this is backing up because of that scarring and we can't get blood flow through scar tissue.

25:26

25 minutes, 26 seconds

Um we've got just tons and tons of blood in here and they're weakwalled so they're very easy to pop open.

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25 minutes, 36 seconds

and burst.

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25 minutes, 38 seconds

So these varaces can cause GI bleeds and hemorrhage.

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25 minutes, 44 seconds

A ruptured esophageal vari is an emergency and most people do not survive a ruptured esophageal veracy.

25:54

25 minutes, 54 seconds

Um other thing it does is it backs up in the spleen. This is

26:01

26 minutes, 1 second

your spleen. Your spleen is responsible for like getting those um red blood

26:08

26 minutes, 8 seconds

cells that are old out of the circulatory system and kind of filters through that stuff. So when the spleen

26:16

26 minutes, 16 seconds

backs up with blood, it it doesn't know which blood cells to get rid of or healize hemolyze.

26:26

26 minutes, 26 seconds

So you just have a lot of hemalization hemolization hemolysis.

26:36

26 minutes, 36 seconds

You have a lot of homalysis of red blood cells. So you have a lot of breakdown of red blood cells which also causes anemia.

26:46

26 minutes, 46 seconds

Let's go back here. Paral scar tissue impairs good blood flow back of a blood portal hypertension esophageal. Okay,

26:54

26 minutes, 54 seconds

yep, got those. And asites. Okay. So the other reason why asites happens is not

27:00

27 minutes

only the um decreased encodic pressure or osmotic pressure

27:09

27 minutes, 9 seconds

but also we've got increased hydrostatic pressure now in these veins in our abdomen

27:16

27 minutes, 16 seconds

which puts us puts us at even more risk for edema or third spacing into that peritineal cavity.